

AVI Planner

Standalone ESAPI AutoPlan tool

Deployment Guide

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Overview

GitHub repository [AVIPlanner](#) contains source code for compiling, configuring, and deploying standalone desktop application **AVI Planner**, which can automatically create Head & Neck VMAT treatment plans, using Varian Eclipse Scripting API (ESAPI, writable version v15 or above).

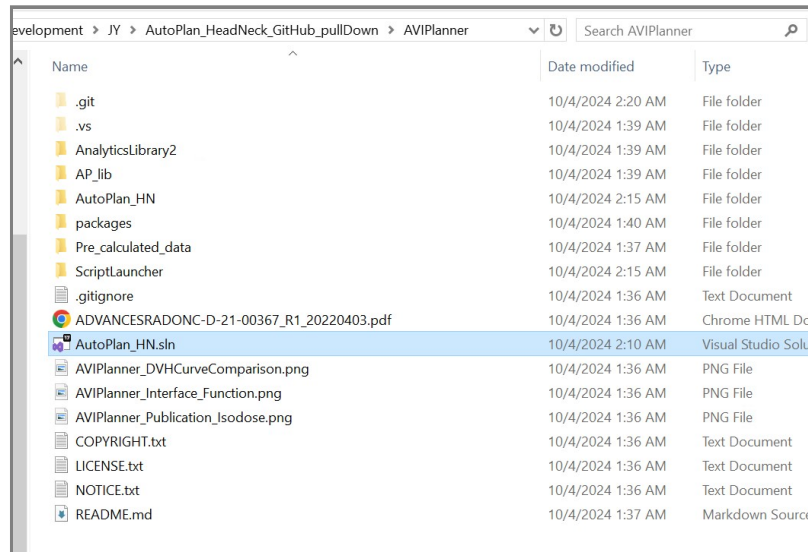
Eclipse Scripting API based Automatic planner for Head and Neck VMAT. The software is solely for non-commercial, non-clinical research and education use in support of the publication.

Create RT plan based on historical manual planning experiences (summarized into pre-calculated Json files).

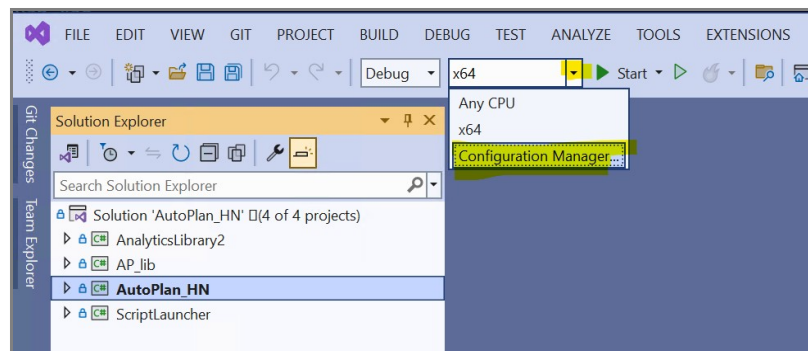
Compile the code

Clone from GitHub repository [AVIPlanner](#) to a local machine where Varian Eclipse treatment planning system is installed (thick Eclipse client).

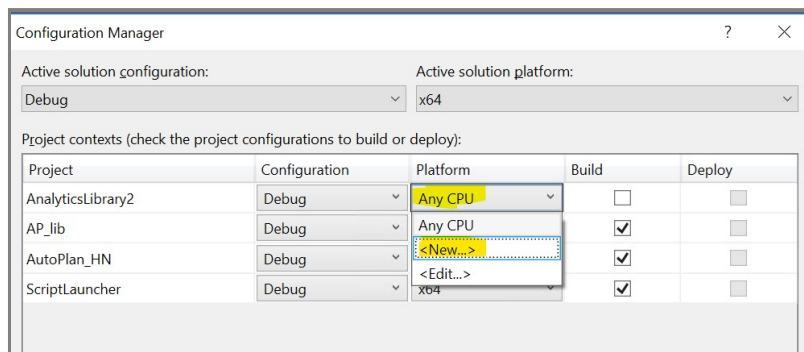
Open the solution file **AutoPlan_HN.sln** with [Visual Studio](#).



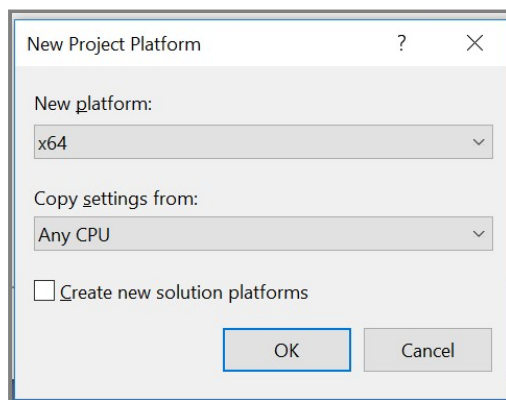
The code needs to be compiled with x64 configuration. Open **Configuration Manager** and make sure each project have **Platform** set as **x64**.



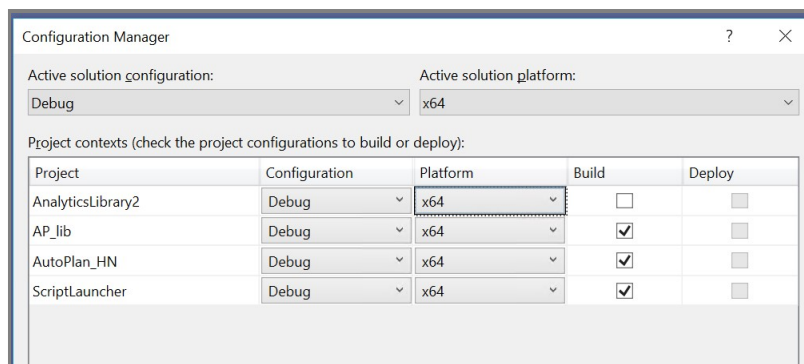
If you don't see **x64** exist in the pulldown list, create it from <New...> option.



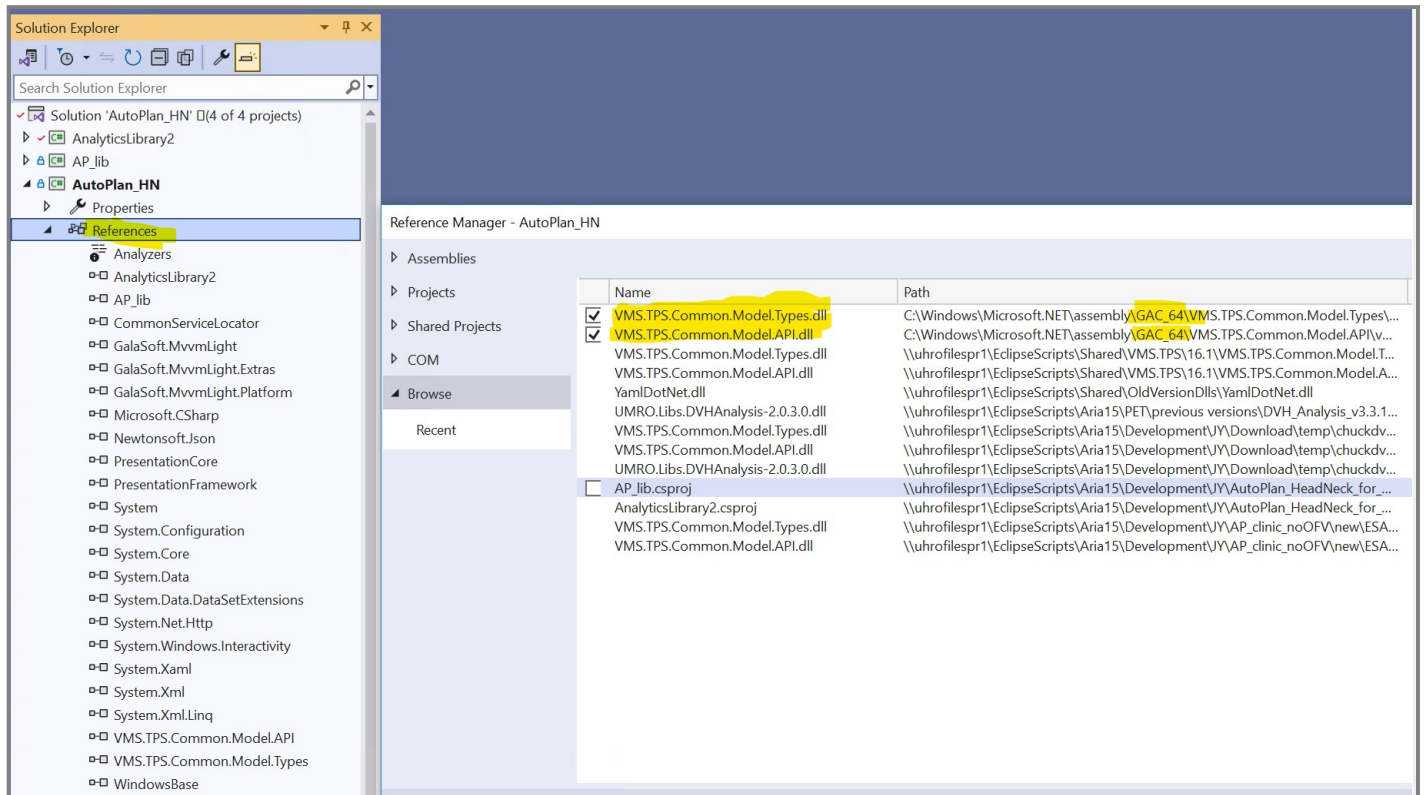
Create x64 platform by 'Copy settings from' Any CPU. Click OK.



Make sure **x64** is selected for all projects in this solution:

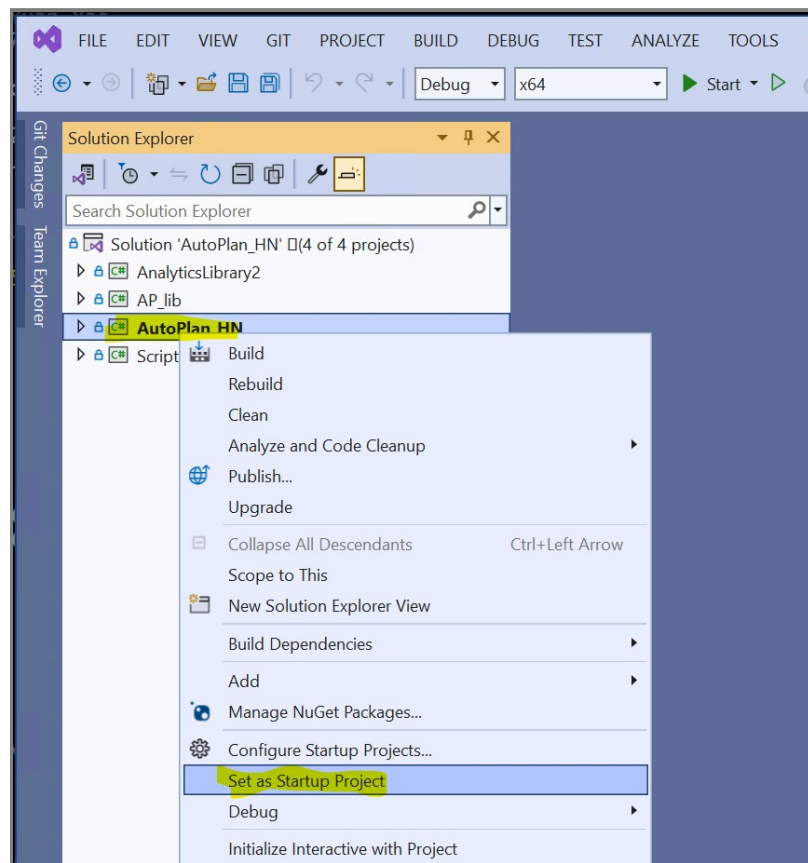


Add reference to two VMS.TPS dll files. (If you are on a thick client where Eclipse is installed locally, this should happen automatically when you build the solution, since the two dlls are already in system GAC. If not, manually locate these two dll on your machine.)

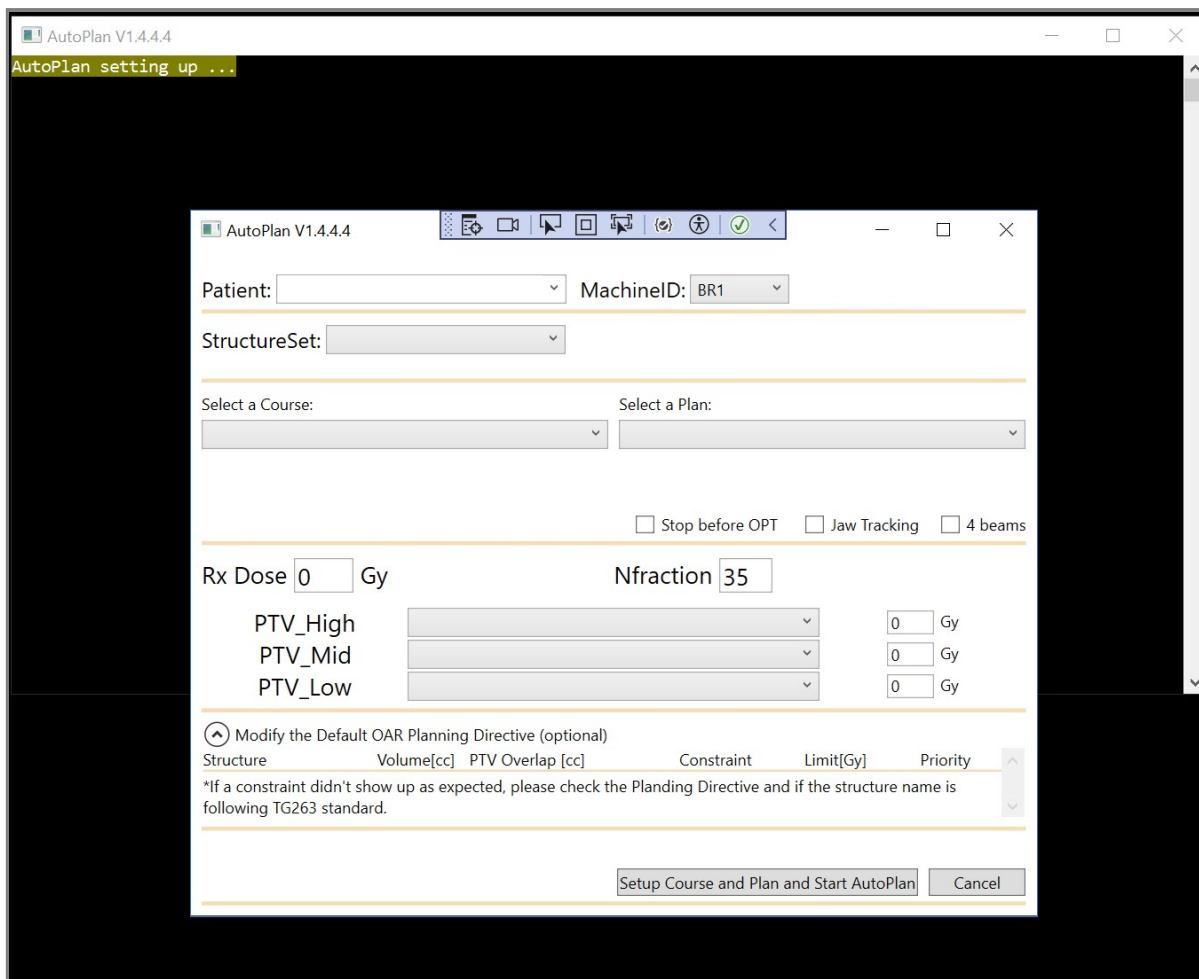


Add the same dll reference for project **AP_lib**.

Set project **AutoPlan_HN** as **startup** project, build the solution and Start. (This only happens on a thick Eclipse client. If not, build the project and see section below about **Script Launcher**)

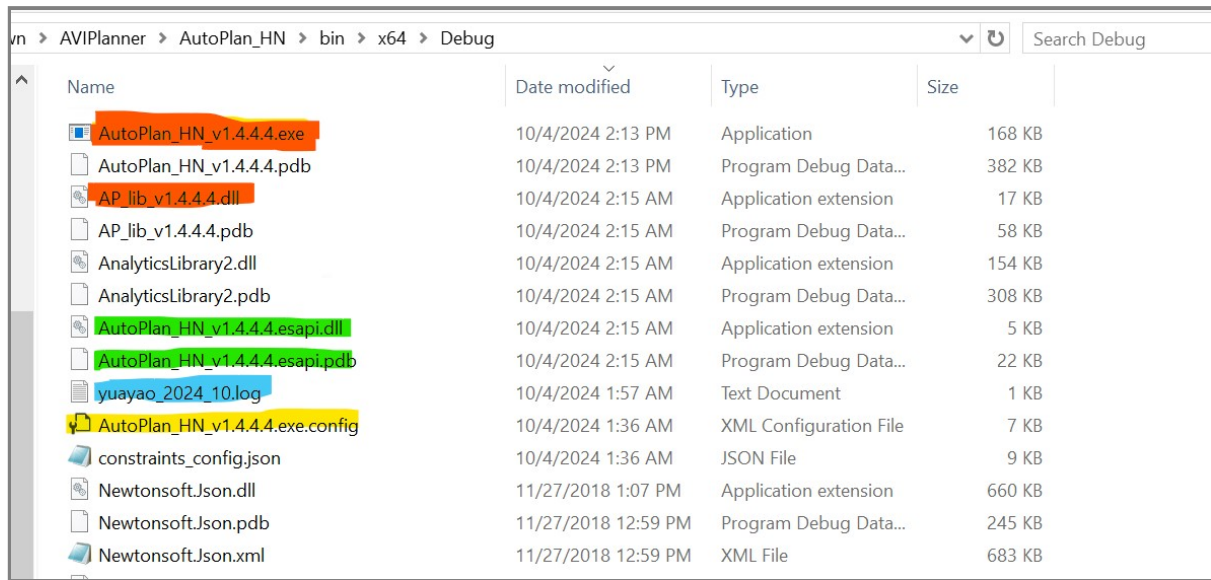


You should be able to see the following console window and UI panel for interaction:



If you build and run successfully. You should be able to see the following files in folder

`\AutoPlan_HN\bin\x64\Debug`:

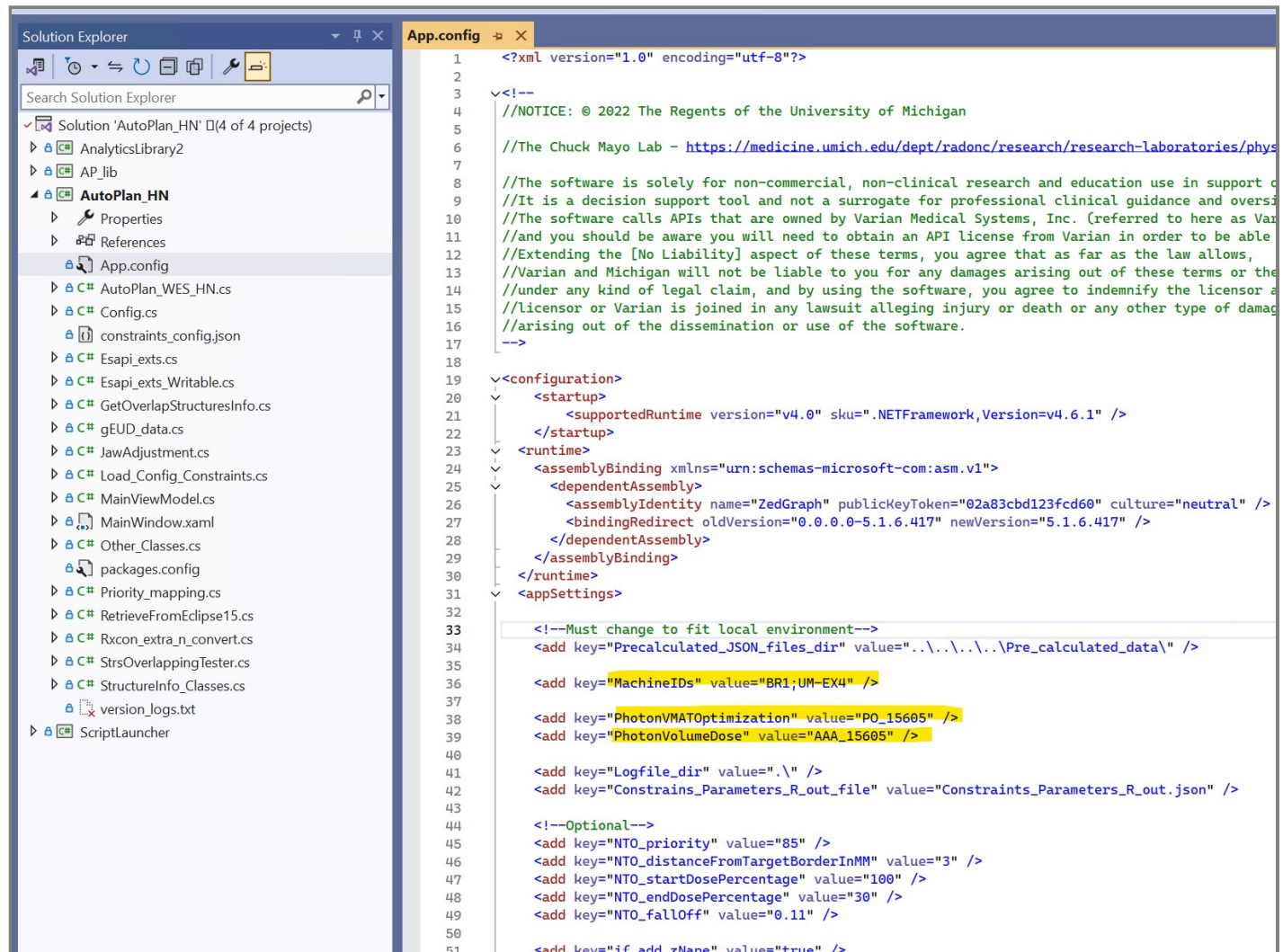


Name	Date modified	Type	Size
AutoPlan_HN_v1.4.4.4.exe	10/4/2024 2:13 PM	Application	168 KB
AutoPlan_HN_v1.4.4.4.pdb	10/4/2024 2:13 PM	Program Debug Data...	382 KB
AP_lib_v1.4.4.4.dll	10/4/2024 2:15 AM	Application extension	17 KB
AP_lib_v1.4.4.4.pdb	10/4/2024 2:15 AM	Program Debug Data...	58 KB
AnalyticsLibrary2.dll	10/4/2024 2:15 AM	Application extension	154 KB
AnalyticsLibrary2.pdb	10/4/2024 2:15 AM	Program Debug Data...	308 KB
AutoPlan_HN_v1.4.4.4.esapi.dll	10/4/2024 2:15 AM	Application extension	5 KB
AutoPlan_HN_v1.4.4.4.esapi.pdb	10/4/2024 2:15 AM	Program Debug Data...	22 KB
yuayao_2024_10.log	10/4/2024 1:57 AM	Text Document	1 KB
AutoPlan_HN_v1.4.4.4.exe.config	10/4/2024 1:36 AM	XML Configuration File	7 KB
constraints_config.json	10/4/2024 1:36 AM	JSON File	9 KB
Newtonsoft.Json.dll	11/27/2018 1:07 PM	Application extension	660 KB
Newtonsoft.Json.pdb	11/27/2018 12:59 PM	Program Debug Data...	245 KB
Newtonsoft.Json.xml	11/27/2018 12:59 PM	XML File	683 KB

- **Red color** - The resulting `AutoPlan_HN_v1.4.4.4.exe` and `AP_lib_v1.4.4.4.dll`. These two files will need to be approved for usage in Clinic Eclipse environment.
- **Yellow color** - Configuration file
- **Blue color** - per user log file (will be created when a user creates an auto plan with `AutoPlan_HN_v....exe`)
- **Green color** - Script Launcher (copied from folder `\ScriptLauncher\bin\x64\Debug`)

Configuration

Configuration file `AutoPlan_HN_v1.4.4.4.exe.config` has a few settings that must be changed to reflect local deployment environment:



Precalculated_JSON_files_dir - path to the folder where pre-computed json files are stored. These json files summarize experience of University of Michigan HN plans from 2016-2019:

AVIPlanner > Pre_calculated_data			
Name	Date modified	Type	Size
Constraints_Parameters_R_out.json	10/4/2024 1:37 AM	JSON File	35
Summary_dict_HN_General_0.json	10/4/2024 1:37 AM	JSON File	3,885
Summary_dict_HN_General_0_StatsDVH.json	10/4/2024 1:37 AM	JSON File	2,640

MachineIDs needs to be changed to the available machines on site. Each available machine ID is separated by a semicolon ;.

Fields	Dose	☐ Field Alignments	☐ Clinical Goals	☐ Optimization Objectives	Dose Statistics	Re	
Group	Field ID	Technique	Machine/Energy	MLC	Field Weight	Scale	G
I	CW	Arc Therapy-I	BR1 - 6X	VMAT	1.084	Varian IEC	18
I	CCW	Arc Therapy-I	BR1 - 6X	VMAT	1.227	Varian IEC	178
I	CW2	Arc Therapy-I	BR1 - 6X	VMAT	0.811	Varian IEC	18

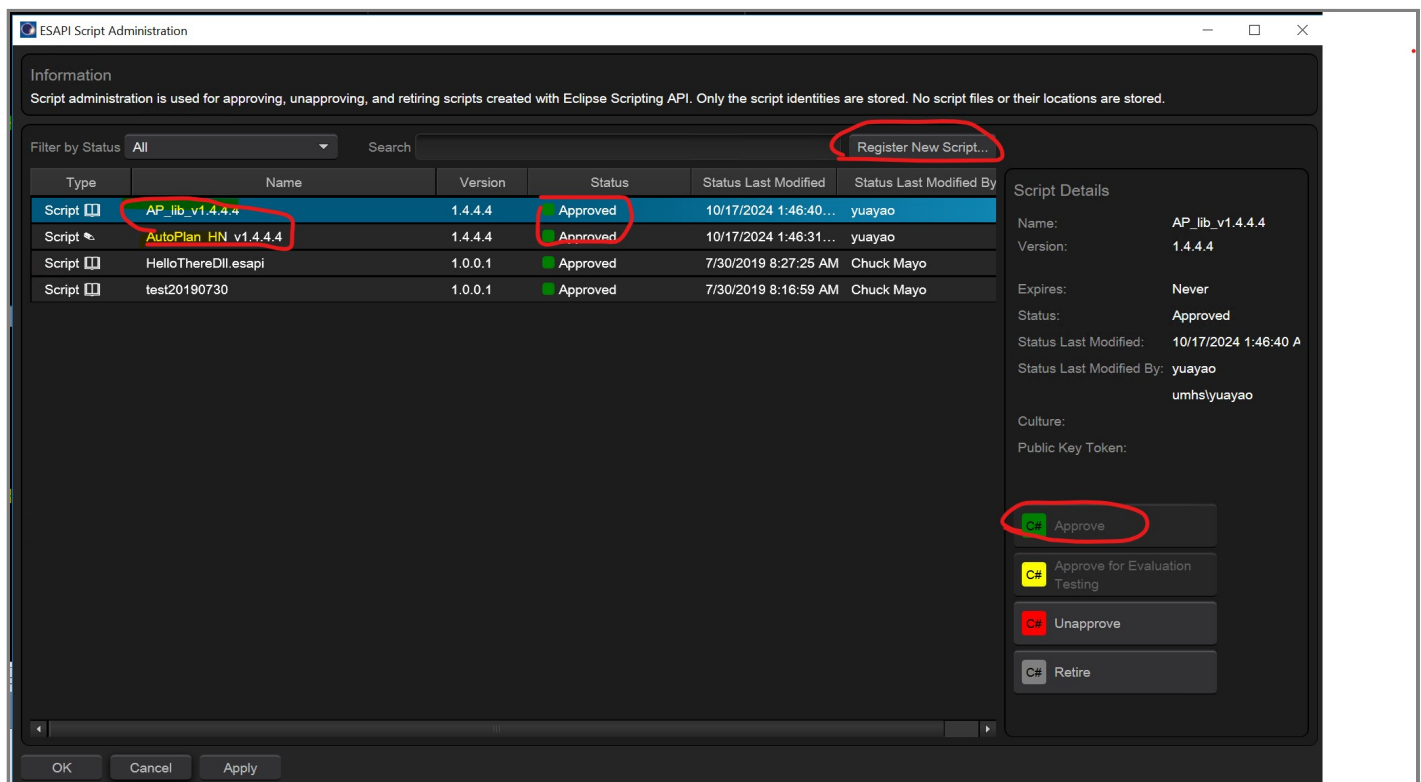
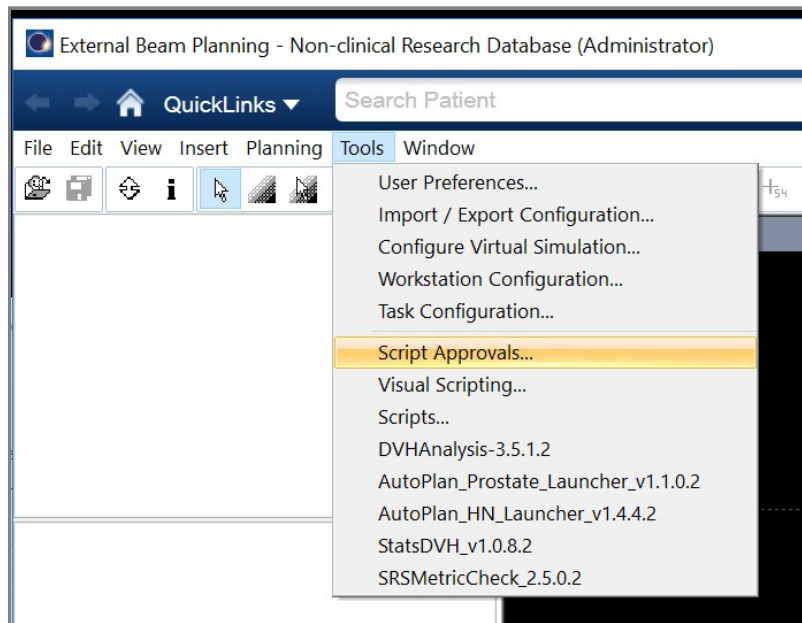
PhotonVMATOptimization and **PhotonVolumeDose** specify the algorithms to use for Optimization and Volume Dose calculation:

Brain ☒ Brainstem ☒ Brainstem_PRV03 ☒ Cochlea_L ☒ Cochlea_R ☒ Esophagus ☒ GlnD_Submand_L <

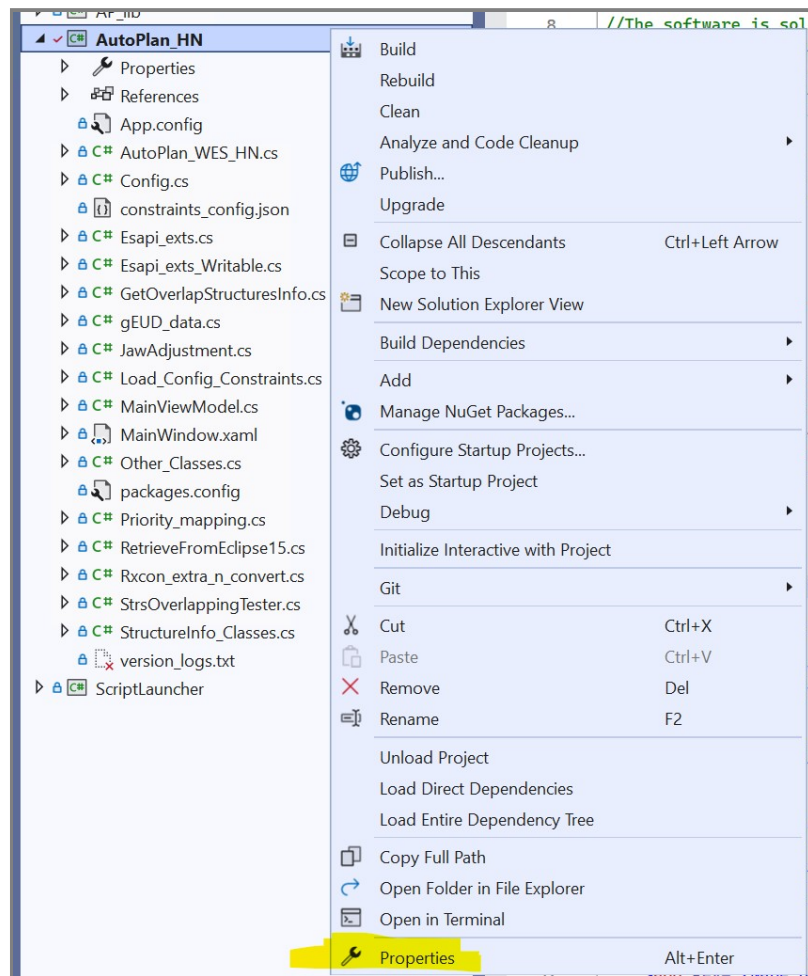
Change the **Logfile_dir** if needed.

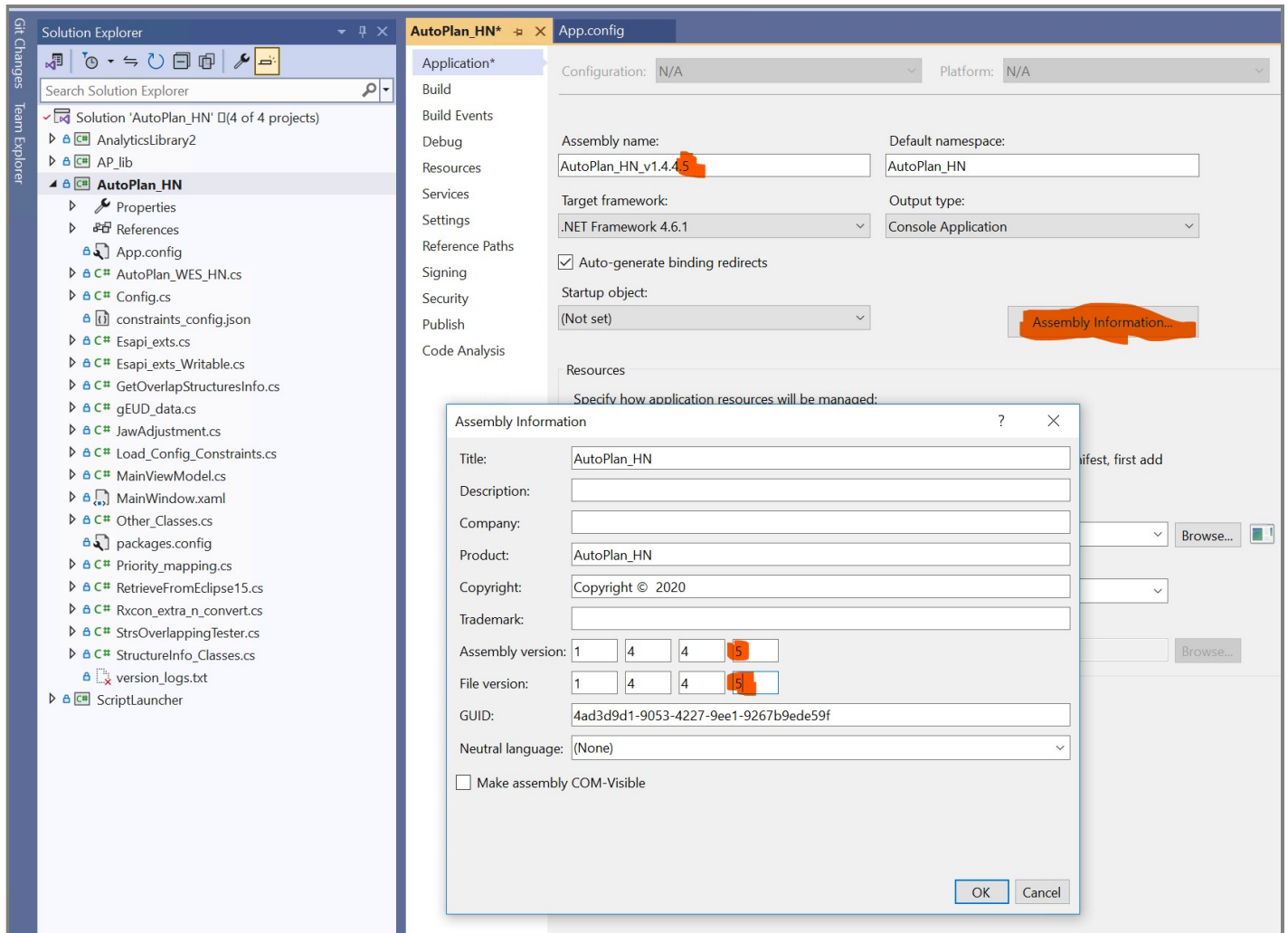
Script Approval

Varian requires writable script be approved before use in Clinical environment. In order to use this AVI Planner program (script) in clinical Eclipse environment, you will need to approve two files: `AutoPlan_HN_v....exe` and `AP_lib_v....dll`



If for some reason you need to re-compile the code. To pass the above approval steps, you need to increase file versions. (we recommend putting version number in file names and increasing them as well). For example, here is how you do it for project **AutoPlan_HN**:





Remember to increase version number for project `AP_lib` and then approve it as well

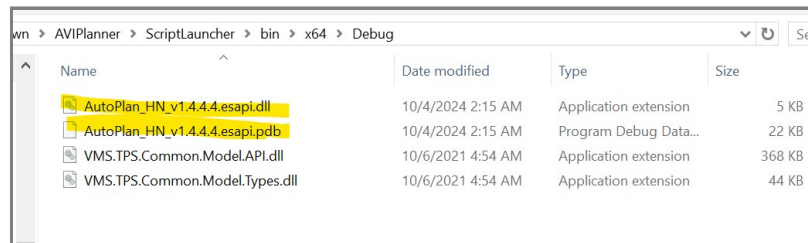
If renaming `AP_lib` 'Assembly name' with new version number breaks the reference between `AutoPlan_HN` and `AP_lib` project in Visual Studio, delete and re-add the reference.

Script Launcher

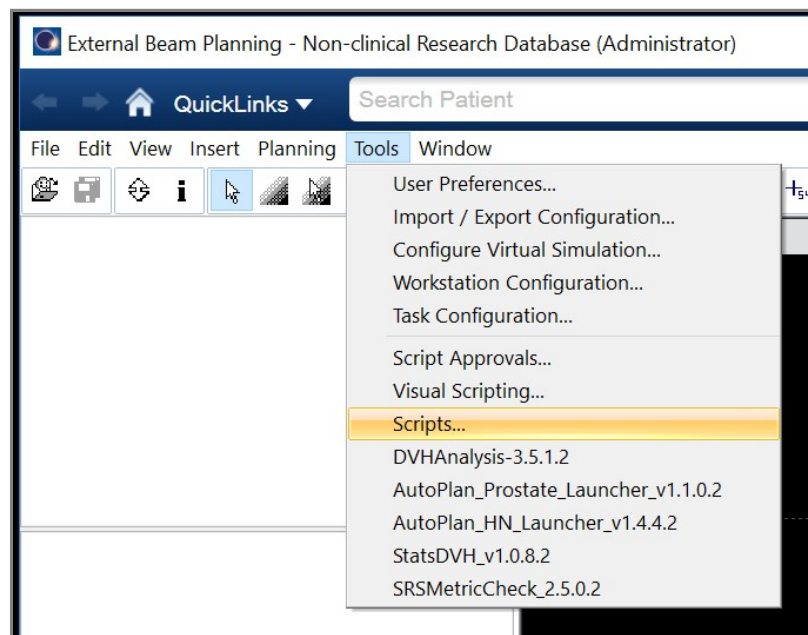
On a thick client where Eclipse is installed locally, user can directly double click **AutoPlan_HN_v1.4.4.4.exe** to run the script.

If a user needs to run from Citrix environment, we need a script launcher to trick Eclipse to load our standalone exe file through a script plugin esapi.dll.

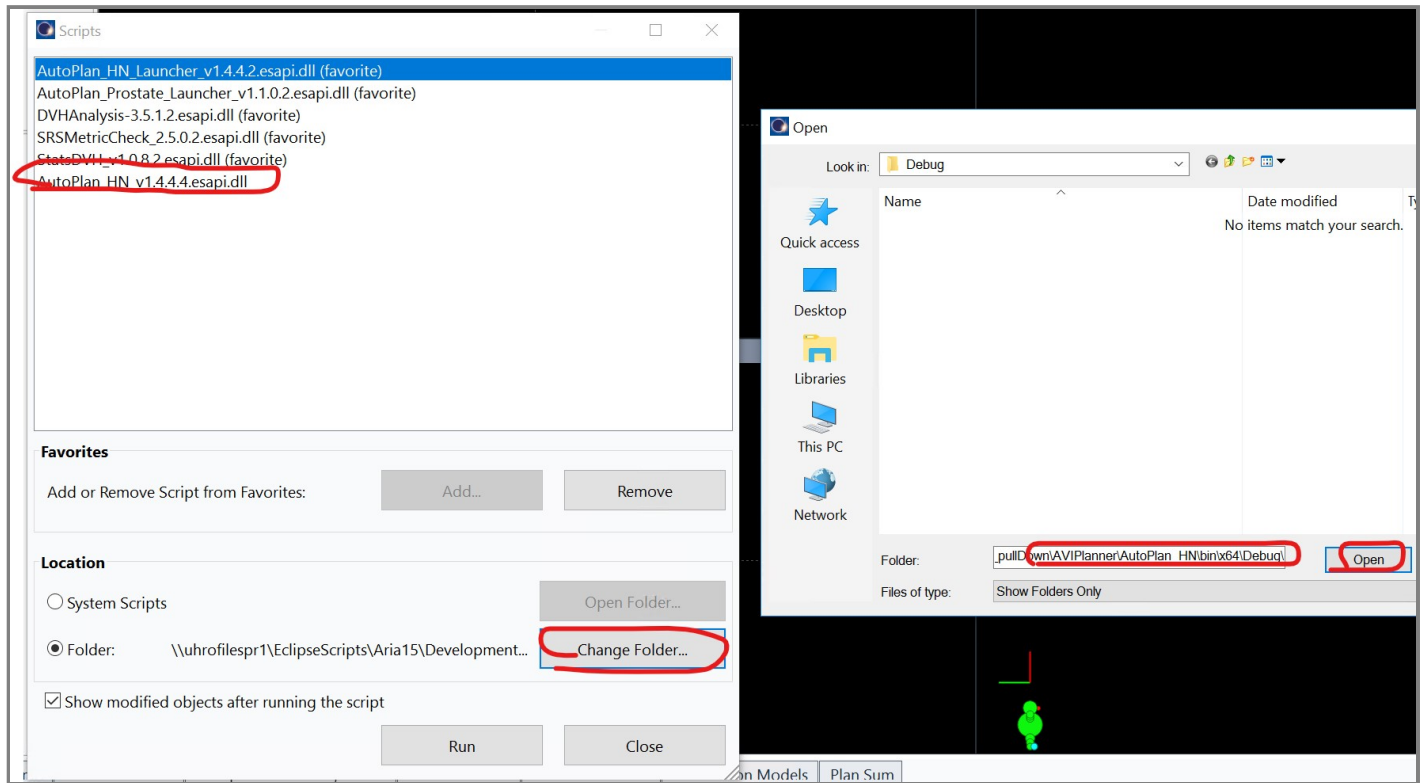
When you build the solution, under project **\ScriptLauncher** bin folder **\bin\x64\Debug**, the launcher files are created. Copy them into the same folder with **AutoPlan_HN_v1.4.4.4.exe**



Next, in Eclipse, go to **Tools -> Scripts...**



Change folder to where `AutoPlan_HN_v1.4.4.4.exe` and `AutoPlan_HN_v1.4.4.4.esapi.dll` are located and open the folder. You should see the launcher `AutoPlan_HN_v1.4.4.4.esapi.dll` show up in the script list, double click to run.



Here, Eclipse Script window only recognizes files ending with `esapi.dll`. Launcher `AutoPlan_HN_v1.4.4.4.esapi.dll` will load our real exe program `AutoPlan_HN_v1.4.4.4.exe` and execute it.

If everything goes well, you should be able to create plans with AVI Planner now:

AutoPlan V1.4.4.4

Patient: [REDACTED] MachineID: BR1

StructureSet: AP_20211222HN

Select a Course: \$AP Select a Plan: create new plan

new plan ID: \$auto

☐ Stop before OPT ☐ Jaw Tracking ☐ 4 beams

Rx Dose 70 Gy Nfraction 35

PTV_High PTV_Mid PTV_Low

PTV_High^70 PTV_Mid^66.5 PTV_Low^59.5

70 Gy 66.5 Gy 59.5 Gy

⬆️ Modify the Default OAR Planning Directive (optional)

Structure	Volume[cc]	PTV Overlap [cc]	Constraint	Limit[Gy]	Priority
Bone_Mandible	77.3	M_3.3 L_10.1	D0.1cc[Gy]	70	3
Brain	1,461.5	L_0.0	Mean[Gy]	60	3
Brainstem	18.6		D0.1cc[Gy]	54	1
Brainstem_PRV03	34.0	L_0.0	D0.1cc[Gy]	54	1.5
Cavity_Oral	90.7	M_2.0 L_67.4	Mean[Gy]	30	4
Cochlea_L	0.1		D0.1cc[Gy]	40	3
Cochlea_R	0.1		D0.1cc[Gy]	40	3
Esophagus	10.1	M_0.3 L_0.4	Mean[Gy]	20	1.5
Larynx	35.3	M_0.1 L_2.0	Mean[Gy]	20	1.5
Lips	20.6		V35Gy[%]	5	2.5
Musc_Constrict_I	8.1	M_0.4 L_0.7	Mean[Gy]	20	1.5
Musc_Constrict_S	10.2	H_1.1 M_0.6 L_9.6	Mean[Gy]	50	4
Parotid_L	23.9	M_1.2 L_0.5	Mean[Gy]	24	1.5
Parotid_R	24.5	L_1.7	Mean[Gy]	24	1.5
SpinalCord	31.1		D0.1cc[Gy]	45	1
SpinalCord_PRV05	103.9		D0.1cc[Gy]	50	1.5

*If a constraint didn't show up as expected, please check the Planning Directive and if the structure name is following TG263 standard.

Setup Course and Plan and Start AutoPlan Cancel

If you experience some error, please consult the log file produced in the same folder first. And contact the developer if needed.